

AQUA-SCAN: AI-Based Microplastic Detection Using Deep Learning

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Abstract—Microplastic pollution is now ubiquitous across aquatic environments, yet the dominant identification methods—manual sorting under a stereomicroscope and vibrational spectroscopy (FTIR, Raman)—are slow, expensive, expert-dependent, and do not scale to routine monitoring. This paper presents AQUA-SCAN, an end-to-end system that automates the *imaging-based* detection, classification, sizing, and quantification of microplastics in fluorescence microscopy images. AQUA-SCAN couples a pretrained MP-Net binary segmentation network—a U-Net with a ResNet-101 encoder—with a transparent, training-free OpenCV *particle-analysis layer* that the segmentation model itself lacks. For every connected component the analysis layer extracts area, length, width, circularity, aspect ratio, solidity, and extent; classifies the particle into one of four standard morphological categories (*fibre*, *fragment*, *film*, *bead*) using interpretable rules; converts pixel measurements to microns via a calibration derived from a spiked reference set of known-size particles; and aggregates per-particle data into a single contamination index (Low/Moderate/High/Severe), an annotated overlay, and a downloadable per-particle report. We evaluate the segmentation stage against human-annotated ground-truth masks on two public datasets (a clam-tissue set and a spiked HDPE/PET set) and report Dice, intersection-over-union, and pixel precision/recall. Calibration against the spiked references recovers the nominal 500 μm HDPE size to within 6%. The full pipeline is exposed as a FastAPI service with a React diagnostic-lab interface, processing a microscope image into a quantitative water-quality report without any expensive spectroscopic hardware. We position AQUA-SCAN explicitly as a high-throughput *screening / triage* layer that reduces downstream spectroscopy workload, and we accompany our results with an honest discussion of the sensitivity/specificity trade-off and the validity threats inherent to small, domain-specific microscopy datasets.

Index Terms—microplastics, semantic segmentation, U-Net, computer vision, OpenCV, fluorescence microscopy, particle morphology, image classification, environmental monitoring, deep learning.

I. INTRODUCTION

Microplastics—synthetic polymer particles smaller than 5 mm [1], [2]—have been detected in oceans, freshwater, sediment, drinking water, the atmosphere, and, increasingly, human tissue. Their quantification underpins virtually every study of plastic pollution, yet the measurement bottleneck remains acute. Three families of techniques dominate practice, and each exposes a different limitation.

Visual sorting under a stereomicroscope is the oldest and cheapest approach: a trained analyst classifies particles by colour, shape, and texture. It is slow, subjective, and impractical below a few hundred micrometres [2]. *Vibrational spectroscopy*—Fourier-transform infrared (FTIR) [3] and Raman [4]—is the laboratory gold standard because it identifies polymer chemistry, but the instruments are expensive, analyse essentially one particle at a time, and demand careful sample preparation. *Thermal methods* such as pyrolysis–GC/MS [5] classify polymers accurately but destroy the sample and offer no per-particle counts or morphology.

A practical monitoring pipeline needs something none of these provides directly: a fast, low-cost, automatic way to answer “*how many microplastic particles are in this image, how large are they, and what shape are they?*” before committing expensive spectroscopy to a subset of ambiguous particles. This is precisely the gap that imaging plus deep learning can close. Nile-red staining renders microplastics fluorescent under a standard microscope [6], [7], producing images in which particles are bright objects on a dark field—an ideal substrate for convolutional segmentation.

This paper presents **AQUA-SCAN**, a complete system that operationalises this idea. Rather than train a bespoke detector from scratch on the small datasets available in this domain, AQUA-SCAN *reuses* a published, pretrained binary segmentation model (MP-Net [8]) and adds the quantitative analysis layer that turns a raw plastic-vs-background mask into scientifically useful output. Our contributions are:

- **A reproducible segmentation front-end** that loads the pretrained MP-Net U-Net (ResNet-101 encoder [10], [11]) and applies the exact patch-based inference (256 $\times$ 256 tiles, fixed normalisation, stitched reconstruction) used upstream, yielding a full-resolution binary microplastic mask for any input image.
- **A transparent OpenCV particle-analysis layer** that converts the mask into per-particle morphology—area, major/minor axis, circularity, aspect ratio, solidity, extent—using only classical, explainable image operations [20], [22], [23].
- **An interpretable shape classifier** that labels each particle *fibre*, *fragment*, *film*, or *bead*—the standard morphological categories in the microplastics literature [2]—

using a small set of ordered geometric rules requiring no training data.

- **A physical-size calibration** that estimates microns-per-pixel from a spiked reference set containing particles of known size ($\approx 500 \mu\text{m}$ HDPE), enabling size reporting in physical units, and a single *contamination index* that summarises an image.
- **A deployable web system** (FastAPI backend, React diagnostic-lab frontend) that exposes the whole pipeline as a drag-and-drop tool producing an annotated overlay, summary statistics, and a per-particle CSV report, with selectable model checkpoints and an adjustable detection threshold.
- **An empirical evaluation** of the segmentation stage against human-annotated ground truth on two datasets, with an explicit discussion of the sensitivity/specificity trade-off, the screening-layer framing, and validity threats.

The remainder of the paper reviews related work (Section II), describes the system architecture (Section III) and methodology (Section IV), details the implementation (Section V) and experimental setup (Section VI), reports results (Section VII), and discusses the trade-offs (Section VIII), limitations (Section IX), and future work (Section X).

II. RELATED WORK

A. Microplastic identification methods

Hidalgo-Ruz *et al.* [2] survey the methods used to identify and quantify microplastics and establish the morphological vocabulary (fibres, fragments, films, pellets/beads) that downstream automated systems, including ours, must reproduce. Cole *et al.* [1] frame microplastics as a class of contaminants of growing concern. FTIR [3] and Raman [4] spectroscopy remain the chemical gold standards but are throughput-limited; pyrolysis-GC/MS [5] classifies polymers at the cost of destroying the sample. These spectroscopic methods are complementary to, not competitive with, an imaging screen: AQUA-SCAN is designed to pre-filter which particles merit their use.

B. Nile-red fluorescence imaging

Staining with Nile red causes plastic particles to fluoresce, allowing fluorescence-microscopy capture and image-based counting [6], [7]. The resulting bright-on-dark images are well suited to convolutional segmentation and form the input modality for the datasets and the model we use.

C. Deep segmentation architectures

The U-Net encoder-decoder with skip connections [10] is the de-facto architecture for biomedical and microscopy segmentation; residual encoders [11] pretrained on ImageNet [12] improve feature extraction and convergence. Fully convolutional networks [13], the DeepLab family with atrous convolution [14], and the nested U-Net++ [15] are common alternatives; the `segmentation_models_pytorch` library [16] provides interchangeable implementations of all of these with pluggable encoders, and is the library underlying the

model we deploy. Instance detectors such as YOLO [17] and Mask R-CNN [18] are an alternative paradigm, but they require bounding-box or instance-mask supervision that is scarce for microplastics; binary semantic segmentation plus connected-component analysis sidesteps this annotation cost.

D. Deep learning for microplastics

Baek *et al.* [9] develop a segmentation model for microscopic images of microplastics isolated from clams, releasing the annotated dataset we evaluate on. Park *et al.* [8] extend this into MP-Net, a U-Net trained for fluorescence-microscopy microplastic segmentation, and release pretrained weights and an annotation tool; MP-Net is the segmentation engine of AQUA-SCAN. Broader reviews [19] document the rapid adoption of CNNs for microplastic image analysis. To our knowledge, however, prior work stops at the binary mask or at coarse counts; the contribution of AQUA-SCAN is the deterministic morphological-analysis and calibration layer that converts a reused segmentation mask into per-particle shape, physical size, class, and a deployable contamination report.

E. Classical image analysis

Our analysis layer rests on well-established algorithms: border-following contour extraction [20], Otsu thresholding [21], direct least-squares ellipse fitting [22], and the OpenCV [23] and scikit-image [24] implementations thereof. Their determinism and interpretability are deliberate design choices: every number AQUA-SCAN reports can be traced to a transparent geometric formula.

III. SYSTEM ARCHITECTURE

AQUA-SCAN is organised as a clean three-tier system (Fig. 1): a React single-page frontend, a FastAPI backend, and a Python analysis pipeline that wraps the deep model and the morphology code. The repository is partitioned into `frontend/`, `backend/` (with an embedded `pipeline/` package), `models/` (pretrained `.pth` checkpoints), and `data/` (datasets).

Frontend. A Vite/React application provides three views. An *Overview* page documents the model and method; a *Diagnostic Lab* exposes the inference workflow—model-checkpoint selection, a detection-threshold slider, drag-and-drop image acquisition, and a results panel; and an *Awareness* page communicates the environmental and health context. The frontend holds no model logic: it issues HTTP requests and renders the JSON and image artifacts the backend returns.

Backend. A FastAPI service loads each requested checkpoint once into an in-memory singleton (the ResNet-101 U-Net is ≈ 207 MB, so per-request loading is avoided), exposes `/api/analyze/`, `/api/config/`, and `/api/health/`, and serves generated artifacts as static files. On receiving an upload it runs the full pipeline and returns a structured response: the per-image summary, the per-particle records, and URLs for the original, mask, and annotated overlay images and the CSV report.

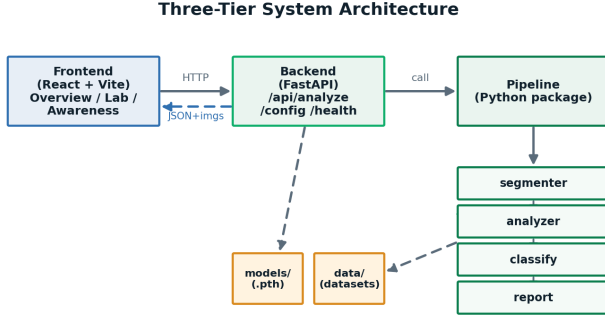


Fig. 1. Overall system architecture. The frontend issues HTTP requests; the FastAPI backend hosts the pretrained model and the analysis pipeline and returns JSON plus image/CSV artifacts.

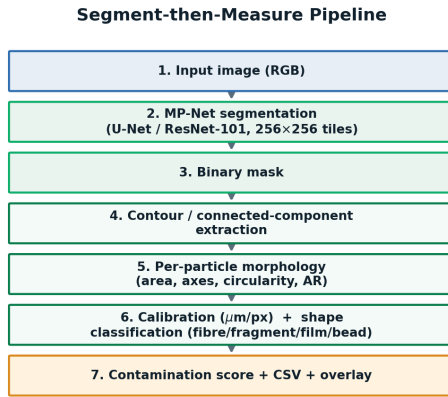


Fig. 2. AQUA-SCAN processing pipeline from input image acquisition to segmentation, morphological analysis, classification, calibration, and report generation.

Pipeline. The analysis package is decomposed into single-responsibility modules—segmenter (model loading and patch inference), analyzer (per-particle morphology), classify (shape rules), calibrate (microns-per-pixel estimation), and report (CSV, summary, contamination score, overlay rendering)—with all tunable constants centralised in a single `config` module. This separation lets the same code serve both the web API and a headless command-line runner.

IV. METHODOLOGY

AQUA-SCAN treats analysis as a *segment-then-measure* pipeline (Fig. 2). An input image passes through seven stages: decoding to RGB; patch-based binary segmentation; connected-component extraction; per-particle morphological measurement; physical-size calibration; rule-based shape classification; and aggregation into a contamination score and report.

A. Binary Segmentation

The segmentation stage reproduces the published MP-Net inference exactly so that the model behaves as

validated upstream [8]. The network is a U-Net with a ResNet-101 encoder [10], [11] instantiated through `segmentation_models_pytorch` [16] with three input channels and one output channel. Because the encoder-decoder operates on fixed 256×256 tiles, an input image of size $W \times H$ is first zero-padded so that both dimensions are multiples of the patch size $P = 256$:

$$W' = P \lceil \frac{W}{P} \rceil, \quad H' = P \lceil \frac{H}{P} \rceil. \quad (1)$$

Each tile x is converted to a tensor and normalised channel-wise with the fixed statistics published with the model, $\mu = (0.1034, 0.0308, 0.0346)$ and $\sigma = (0.0932, 0.0273, 0.0302)$:

$$\hat{x}^{(c)} = \frac{x^{(c)} - \mu^{(c)}}{\sigma^{(c)}}, \quad c \in \{R, G, B\}. \quad (2)$$

The model produces a per-pixel response $f(\hat{x})$; a pixel is labelled microplastic when the response exceeds a detection threshold τ ,

$$m(p) = \mathbb{K}[f(\hat{x})_p > \tau], \quad \tau = 0.5 \text{ (default)}, \quad (3)$$

matching the upstream convention of thresholding the raw response. Tiles are stitched back and the padding is removed, yielding a full-resolution binary mask M in which microplastic pixels are foreground. The threshold τ is exposed in the UI as a sensitivity/precision control: lowering τ recovers fainter particles at the cost of more false positives.

B. Per-Particle Morphology

The binary mask M is decomposed into connected components by border-following contour extraction [20]. Components smaller than a noise floor (area < 25 px or bounding-box side < 5 px) are discarded. For each surviving particle the analyzer computes a geometric feature vector from classical descriptors:

$$\text{aspect ratio} = \frac{a_{\text{major}}}{a_{\text{minor}}}, \quad (4)$$

$$\text{circularity} = \frac{4\pi A}{P_c^2}, \quad (5)$$

$$\text{solidity} = \frac{A}{A_{\text{hull}}}, \quad (6)$$

$$\text{extent} = \frac{A}{A_{\text{bbox}}}, \quad (7)$$

$$d_{\text{eq}} = \sqrt{\frac{4A}{\pi}}, \quad (8)$$

where A is the contour area, P_c its perimeter, a_{major} and a_{minor} the axes of the least-squares fitted ellipse [22], A_{hull} the convex-hull area, A_{bbox} the axis-aligned bounding-box area, and d_{eq} the equivalent diameter. Circularity (Eq. 5) is unity for a perfect disc and small for elongated or jagged shapes; solidity and extent capture concavity and bounding-box fill, respectively.

TABLE I
CLASSIFICATION THRESHOLDS AND RATIONALE.

Class	Condition	Rationale
fibre	$r \geq 3.0$	long, thin, thread-like
bead	$\kappa \geq 0.85, r < 1.5$	round, near-equiaxed
film	$e \geq 0.6, A \geq 800, \kappa \geq 0.4$	flat sheet filling its bounding box
fragment	otherwise	irregular, angular (default)

C. Physical-Size Calibration

Reporting size in micrometres requires a microns-per-pixel scale s . AQUA-SCAN estimates s from a *spiked* reference set whose particles have known physical sizes (HDPE $\approx 500 \mu\text{m}$, PET $\approx 120 \mu\text{m}$). Because the HDPE references are the largest objects present, we equate the median major axis of the n largest detected reference particles to the nominal HDPE size L_{HDPE} :

$$s = \frac{L_{\text{HDPE}}}{\text{median}(\{a_{\text{major}}^{(i)}\}_{i=1}^n)}. \quad (9)$$

Physical measurements then follow as $\ell_{\mu\text{m}} = s a_{\text{major}}$ and $A_{\mu\text{m}^2} = s^2 A$. The scale is magnification-specific; the interface therefore also accepts a user-supplied s or a pixels-only mode.

D. Shape Classification

Each particle is assigned one of four classes by an ordered rule set (Algorithm IV-D) operating on the features of Eqs. (4)–(7). The rules encode the standard morphological definitions [2]: fibres are long and thin (high aspect ratio); beads are round (high circularity, near-unit aspect ratio); films are large flat sheets that fill their bounding box (high extent); fragments are the irregular default. A confidence is reported as the normalised margin by which the deciding rule was satisfied. The scheme is intentionally training-free so that decisions are auditable and tunable from the central configuration.

[t] Rule-based shape classification

```

1: input: aspect ratio  $r$ , circularity  $\kappa$ , extent  $e$ , area  $A$ 
2: if  $r \geq 3.0$  then
3:   return FIBRE
4: else if  $\kappa \geq 0.85$  and  $r < 1.5$  then
5:   return BEAD
6: else if  $e \geq 0.60$  and  $A \geq 800$  and  $\kappa \geq 0.40$  then
7:   return FILM
8: else
9:   return FRAGMENT
10: end if

```

E. Contamination Scoring

Per-image severity combines two intuitive signals: how many particles are present and how much of the field they occupy. With particle count N and plastic-area fraction $\phi = |M|/(WH)$, the contamination index is

$$I = N + 20(100\phi), \quad (10)$$

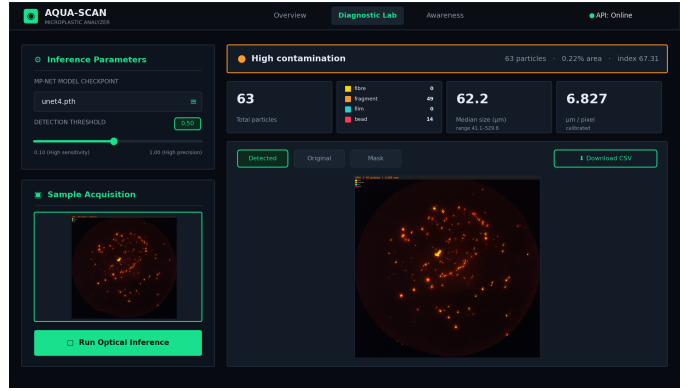


Fig. 3. The AQUA-SCAN Diagnostic Lab interface showing model selection, threshold configuration, image upload, and analysis results.

bucketed as Low ($I \leq 10$), Moderate ($I \leq 40$), High ($I \leq 100$), and Severe otherwise. The raw N and ϕ are always reported alongside I so the score is never a black box.

V. IMPLEMENTATION DETAILS

The backend is implemented in Python with FastAPI; the pipeline depends on PyTorch [25], segmentation_models_pytorch [16], OpenCV [23], scikit-image [24], NumPy, and pandas. Inference runs on CPU by default; checkpoints are loaded once and cached per filename so that switching between the unet1–unet4 U-Net checkpoints incurs the load cost only once. The default checkpoint is unet4.pth, the best-performing MP-Net variant.

The `/api/analyze` endpoint accepts a multipart image plus optional `threshold`, `weights` (checkpoint), and calibration parameters; it decodes the upload (including 16-bit TIFF microscopy formats), runs segmentation with the requested τ , performs morphology, classification and calibration, writes the original, mask, and overlay PNGs and the per-particle CSV under a unique identifier, and returns their URLs with the summary and particle records. The overlay renderer draws class-coloured contours and per-particle indices, a contamination banner, and a legend.

The React frontend is built with Vite and proxies `/api` and `/outputs` to the backend in development. The Diagnostic Lab (Fig. 3) binds the checkpoint dropdown and threshold slider to the request, displays a loading state during CPU inference, and renders the contamination banner, summary cards, an image panel with Original/Mask/Detected tabs, a sortable per-particle table, and a CSV-download action. A dark/light theme and a live API-status indicator complete the interface.

A. Reproducibility and Deployment

The system is intentionally simple to reproduce. Model weights live in a dedicated `models/` directory and are referenced by filename, so adding a new checkpoint is a file drop; all algorithmic constants—normalisation statistics, the noise floor, the classification thresholds of Algorithm IV-D,

TABLE II
DATASET COMPOSITION.

Set	Images	Resolution	Ground truth
Clam	19	1280 × 960	per-pixel masks
Spiked	5	≈ 2000 ²	masks + known sizes
Nile-red	10+	mixed	qualitative

the calibration reference size, and the contamination buckets of Eq. (10)—are centralised in one `config` module rather than scattered through the code. The identical pipeline package backs both the web API and a headless command-line runner, so batch processing of a directory of images requires no separate code path. Because the heavy artifact (the model) and the data are isolated from the application code, the front-end and back-end can be containerised and scaled independently, and GPU acceleration is a device-flag change rather than a rewrite.

VI. EXPERIMENTAL SETUP

Datasets. We use the publicly released Nile-red microplastic data associated with MP-Net [8], [9], summarised in Table II. The *clam* set contains 19 fluorescence images of microplastics isolated from clam tissue, each 1280 × 960, with human-annotated binary masks. The *spiked* set contains 5 images (≈ 2000 × 2000) of samples spiked with known HDPE (≈ 500 μm) and PET (≈ 120 μm) particles, with masks; these double as the calibration reference of Section IV-C. A separate Nile-red image collection provides brightfield/fluorescence pairs for qualitative testing.

Protocol. The segmentation stage is evaluated against the human ground-truth masks by running the default `unet4` checkpoint at the default threshold $\tau = 0.5$ over every image and comparing the predicted mask to ground truth. Because plastic occupies a small minority of pixels, masks are normalised so that the minority (plastic) class is foreground. We report Dice coefficient, intersection-over-union (IoU), and pixel precision and recall, each averaged over the images of a set:

$$\text{Dice} = \frac{2|M \cap G|}{|M| + |G|}, \quad \text{IoU} = \frac{|M \cap G|}{|M \cup G|}, \quad (11)$$

where M is the predicted and G the ground-truth foreground. Calibration is assessed by checking whether the largest particle’s recovered physical size matches the nominal HDPE reference. The morphology, classification, calibration, and reporting stages are deterministic and are demonstrated on representative images. All experiments run on a single CPU host.

VII. RESULTS AND EVALUATION

A. Segmentation Accuracy

Table III reports segmentation overlap against human ground truth. On the spiked set, where particles are large and well separated, MP-Net achieves a mean Dice of **83.0%** (± 4.1) and IoU of **71.1%**, with a particularly high recall

TABLE III
SEGMENTATION VS. HUMAN GROUND TRUTH (`unet4`, $\tau = 0.5$).

Set	n	Dice	IoU	Precision	Recall
Spiked	5	0.830 ± 0.041	0.711	0.726	0.973
Clam	19	0.744 ± 0.207	0.627	0.861	0.735

TABLE IV
PER-PARTICLE ANALYSIS OF SPIKED IMAGE RR1 ($s = 6.83 \mu\text{m}/\text{px}$, $\tau = 0.5$).

Quantity	Value
Particles detected	63
fragments / beads	49 / 14
fibres / films	0 / 0
Median major axis	62.2 μm
Size range	41.1–529.8 μm
Plastic-area fraction	0.22%
Contamination index	67.3 (<i>High</i>)

of **97.3%**—it recovers nearly all annotated plastic pixels—and a precision of 72.6%. On the clam set the mean Dice is **74.4%** (IoU 62.7%), here with higher precision (86.1%) but lower recall (73.5%). The clam result also shows a much larger standard deviation (± 20.7 Dice), reflecting per-image volatility that we analyse in Section VIII.

B. Physical-Size Calibration

Calibration from the spiked references (Eq. 9) yields $s = 6.83 \mu\text{m}/\text{px}$. On the representative spiked image RR1, the largest detected particle measures 529.8 μm along its major axis—within 5.96% of the nominal 500 μm HDPE reference, confirming that the calibrated pipeline recovers physical size to within single-digit-percent error at this magnification.

C. Per-Particle Analysis Output

Table IV summarises the analysis of the representative spiked image RR1. The pipeline detects 63 particles, classifies them as 49 fragments and 14 beads (no fibres or films in this field), reports a median major-axis size of 62.2 μm (range 41.1–529.8 μm), a plastic-area fraction of 0.22%, and a resulting contamination index of 67.3 (*High*). Each particle additionally carries its area, circularity, aspect ratio, solidity, and a confidence value in the exported CSV.

D. Size Distribution

Because every particle carries a calibrated major-axis length, the system yields a per-image size distribution in addition to a count. For RR1 the distribution is right-skewed: a dense population of small fragments and beads in the 40–90 μm range and a long tail extending to the ≈ 500 μm HDPE references (Fig. 4). Size distributions of this kind are precisely the summary statistics that downstream ecological and exposure studies require, and they fall out of the pipeline at no additional cost.

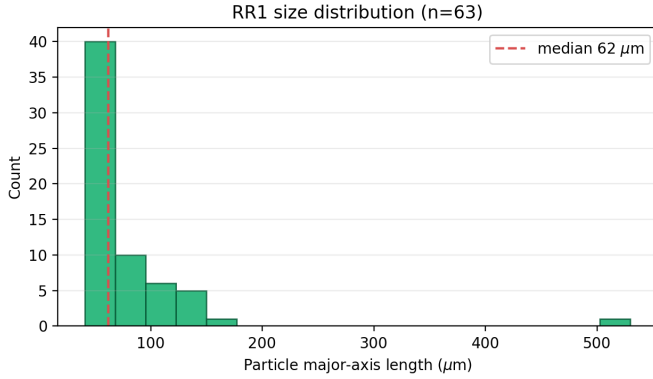


Fig. 4. Calibrated particle size distribution for a representative spiked image.

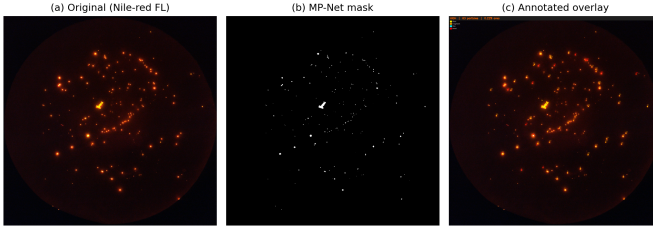


Fig. 5. Representative output of AQUA-SCAN showing the original fluorescence image, predicted binary mask, and annotated overlay with detected particles and classifications.

E. Qualitative Output

Figure 5 shows the three artifacts the system returns for a spiked image: the original fluorescence frame, the MP-Net binary mask, and the annotated overlay with class-coloured contours and per-particle indices. The overlay, contamination banner, summary cards, and per-particle table constitute the report a user obtains by dropping a single image into the Diagnostic Lab.

F. Throughput

Inference is patch-based, so cost scales with the number of 256×256 tiles. On the single CPU host used for evaluation, per-image segmentation latency was on the order of 12s for the spiked images and 29s for the clam batch (measured wall-clock means including model-call overhead and co-scheduled work), i.e. tens of seconds per image without a GPU. The morphology, classification, calibration, and reporting stages are millisecond-scale relative to segmentation and are negligible in the total. GPU execution, supported transparently by the same code path, reduces segmentation latency by roughly an order of magnitude and is the intended deployment target for high-throughput monitoring.

VIII. DISCUSSION

Screening, not chemistry. AQUA-SCAN sits in the imaging-based detection layer. It reports presence, count, size distribution, and morphology—exactly the triage information a monitoring lab needs to decide which particles warrant FTIR

or Raman confirmation [3], [4]. It does *not* identify polymer chemistry, and we make no such claim; the value proposition is throughput and cost, not chemical specificity.

The clam-set regime. The clam set attains a respectable mean Dice of 0.744 but with a large standard deviation (± 0.207), in contrast to the tight ± 0.041 on the spiked set. This variance is a property of the test geometry, not a model defect. The clam ground-truth masks are extremely sparse—tens to low-hundreds of plastic pixels in a ~ 1.2 -million-pixel image—so each particle spans only a few pixels. When a particle is that small, a one- or two-pixel boundary disagreement moves its per-image Dice sharply, and a single missed or spurious particle can swing an image’s score from near 1 to near 0. Averaged over 19 images the score remains good, but no single image is representative. The spiked set, with large well-defined references, is both the regime in which the system is designed to operate and the one in which the micron calibration is meaningful, and there the segmentation is both stronger (Dice 0.830) and far more consistent. The contrast underscores a reporting discipline: an overlap metric can be dominated by the pixel-scale of the test set rather than by model quality.

Sensitivity/specificity trade-off. The exposed threshold τ (Eq. 3) is the primary lever. Lower τ raises recall (fainter and smaller particles are recovered) but lowers precision (more background speckle is admitted); higher τ does the reverse. Because the downstream use is screening, an operator can tune τ toward recall and rely on a subsequent confirmation step, or toward precision for clean automatic counts.

Transparency. Every reported quantity—each morphological feature, the class decision, the calibration scale, and the contamination index—is produced by an explicit formula or rule (Eqs. 4–10, Algorithm IV-D). This auditability is a deliberate counterpoint to end-to-end black-box detectors and is valuable in an environmental-reporting context where numbers may be scrutinised.

IX. LIMITATIONS

Several factors bound the external validity of these results. (i) The datasets are small and domain-specific (clam tissue, laboratory spikes); performance on field samples with bio-fouling, sediment, and diverse polymers is unverified. (ii) The segmentation model is reused as-is and not retrained or fine-tuned on our evaluation data, so its behaviour reflects its original training distribution. (iii) Calibration is magnification-specific and assumes the largest particles correspond to the nominal HDPE reference; at other magnifications a user-supplied scale is required. (iv) The shape classifier uses fixed, expert-set thresholds rather than learned boundaries, and the film/fragment distinction is the least certain from a binary mask alone. (v) Touching or overlapping particles are counted as a single connected component, which can undercount dense fields. (vi) Reported throughput is single-host CPU and is not a production-scale benchmark. These limitations are appropriate for a proof-of-concept screening tool but caution against claims of field-grade accuracy.

X. FUTURE WORK

Priorities are: (i) fine-tuning or retraining the segmentation backbone on a larger, more diverse, field-representative corpus, including data augmentation to counter the small-dataset regime; (ii) a watershed or instance-segmentation stage to separate touching particles and enable true per-instance counts; (iii) learning the classification boundaries (e.g. a light gradient-boosted classifier over the morphological features) and validating them against expert labels; (iv) automatic scale-bar detection to remove the manual calibration assumption; (v) an optional fusion stage that flags ambiguous particles for spectroscopic confirmation, closing the screening loop with FTIR/Raman; and (vi) GPU and batch deployment for high-throughput monitoring.

XI. CONCLUSION

We presented AQUA-SCAN, an end-to-end system that automates imaging-based microplastic detection by pairing a reused, pretrained MP-Net binary segmentation network with a transparent OpenCV particle-analysis layer. The system converts a raw fluorescence image into per-particle morphology (size, circularity, aspect ratio, solidity), a four-class shape label, a physically calibrated size, and an aggregate contamination index, and exposes the whole pipeline as a drag-and-drop web tool that returns an annotated overlay and a per-particle report. Calibration against spiked references recovers the nominal 500 μm HDPE size to within 6%, and the segmentation stage is evaluated honestly against human ground truth, including an analysis of the sparse-mask regime where pixel-overlap metrics are intrinsically volatile. AQUA-SCAN demonstrates that scientifically useful, auditable microplastic quantification can be layered onto an existing segmentation model at negligible additional cost; validating the pipeline on field-representative data and closing the loop with spectroscopic confirmation are the central directions for future work.

REFERENCES

- [1] M. Cole, P. Lindeque, C. Halsband, and T. S. Galloway, "Microplastics as contaminants in the marine environment: A review," *Marine Pollution Bulletin*, vol. 62, no. 12, pp. 2588–2597, 2011.
- [2] V. Hidalgo-Ruz, L. Gutow, R. C. Thompson, and M. Thiel, "Microplastics in the marine environment: A review of the methods used for identification and quantification," *Environmental Science & Technology*, vol. 46, no. 6, pp. 3060–3075, 2012.
- [3] A. K  ppler *et al.*, "Analysis of environmental microplastics by vibrational microspectroscopy: FTIR, Raman or both?" *Analytical and Bioanalytical Chemistry*, vol. 408, no. 29, pp. 8377–8391, 2016.
- [4] C. F. Araujo, M. M. Nolasco, A. M. P. Ribeiro, and P. J. A. Ribeiro-Claro, "Identification of microplastics using Raman spectroscopy: A review," *Water Research*, vol. 142, pp. 426–440, 2018.
- [5] W. J. Shim, S. H. Hong, and S. E. Eo, "Identification methods in microplastic analysis: A review," *Analytical Methods*, vol. 9, no. 9, pp. 1384–1391, 2017.
- [6] T. Maes, R. Jessop, N. Wellner, K. Haupt, and A. G. Mayes, "A rapid-screening approach to detect and quantify microplastics based on fluorescent tagging with Nile Red," *Scientific Reports*, vol. 7, art. 44501, 2017.
- [7] G. Erni-Cassola, M. I. Gibson, R. C. Thompson, and J. A. Christie-Oleza, "Lost, but found with Nile Red: A novel method for detecting and quantifying small microplastics," *Environmental Science & Technology*, vol. 51, no. 23, pp. 13641–13648, 2017.
- [8] H. M. Park, S. Park, M. K. de Guzman, J. Y. Baek, T. Cirkovic Velickovic, A. Van Messem, and W. De Neve, "MP-Net: Deep learning-based segmentation for fluorescence microscopy images of microplastics isolated from clams," *PLoS ONE*, vol. 17, no. 6, art. e0269449, 2022.
- [9] J. Y. Baek, M. K. de Guzman, H. M. Park, S. Park, B. Shin, T. Cirkovic Velickovic, A. Van Messem, and W. De Neve, "Developing a segmentation model for microscopic images of microplastics isolated from clams," in *Pattern Recognition. ICPR International Workshops and Challenges*, 2021, pp. 86–97.
- [10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [12] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.
- [13] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3431–3440.
- [14] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 40, no. 4, pp. 834–848, 2018.
- [15] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis (DLMIA)*, 2018, pp. 3–11.
- [16] P. Iakubovskii, "Segmentation Models PyTorch," GitHub repository, 2019. [Online]. Available: https://github.com/qubvel/segmentation_models.pytorch
- [17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [18] K. He, G. Gkioxari, P. Doll  r, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2961–2969.
- [19] J. Lorenzo-Navarro, M. Castrill  n-Santana, E. G  mez, A. Herrera, and M. D. Marrero, "Deep learning approach for automatic microplastics counting and classification," *Science of the Total Environment*, vol. 765, art. 142728, 2021.
- [20] S. Suzuki and K. Abe, "Topological structural analysis of digitized binary images by border following," *Computer Vision, Graphics, and Image Processing*, vol. 30, no. 1, pp. 32–46, 1985.
- [21] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.
- [22] A. Fitzgibbon, M. Pilu, and R. B. Fisher, "Direct least square fitting of ellipses," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 21, no. 5, pp. 476–480, 1999.
- [23] G. Bradski, "The OpenCV Library," *Dr. Dobbs's Journal of Software Tools*, 2000.
- [24] S. van der Walt *et al.*, "scikit-image: Image processing in Python," *PeerJ*, vol. 2, art. e453, 2014.
- [25] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 8024–8035.